

RESISTANCE PROFILE AND MALDI-TOF MS IDENTIFICATION OF THE EMERGING PATHOGEN

TRICHOPHYTON INDOTINEAE

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Trichophyton indotineae

- Epidemic of tinea infections in (North) India – mainly *tinea corporis* and *tinea cruris*
- High level of resistance to **terbinafine** (TRB)



Trichophyton indotineae sp. nov.: A New Highly Terbinafine-Resistant Anthropophilic Dermatophyte Species

Rui Kano · Utako Kimura · Maki Kakurai · Junichiro Hiruma · Hiroshi Kamata · Yasushi Suga · Kazutoshi Harada

Trichophyton indotineae - An Emerging
Dermatophyte in India
Perspective

Trichophyton mentagrophytes ITS genotype VIII → *Trichophyton indotineae*

Terbinafine resistance

- Possible consequence of excessive use of over-the-counter antifungal creams and topical steroids or glucocorticoids
- Point mutations in the squalene epoxidase-encoding gene (***SQLE***)



amino acid substitutions:

Phe397Leu (F397L)

Leu393Phe (L393F)

Leu393Ser (L393S)

Ala448Thr (A448T)

TRB resistance

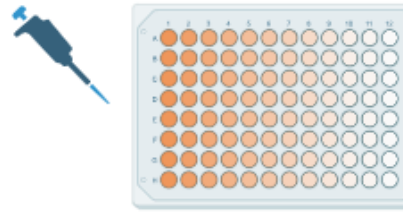
reduced azole
sensitivity

Research

GOAL

Facilitating **diagnosis** and **treatment** of infections caused by *T. indotineae* through:

- The investigation of the **resistance profile** of *T. indotineae* strains
- The evaluation of **MALDI-TOF MS** as an identification tool for these strains



Performing **antifungal susceptibility testing** (AST, following the EUCAST method) to check for possible multi-resistance

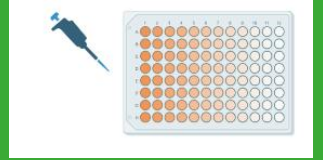


DNA sequencing of the squalene epoxidase-encoding gene (*SQLE*) to investigate present mutations



Validating **MALDI-TOF MS** for rapid and correct identification of *T. indotineae* strains

Antifungal susceptibility testing



- **EUCAST** broth microdilution antifungal susceptibility testing
- Minimal inhibitory concentrations at 100% inhibition of fungal growth (MIC₁₀₀):

fluconazole	terbinafine
itraconazole	griseofulvin
voriconazole	ciclopirox olamine
ketoconazole	naftifine
	amorolfine



EUCAST

EUROPEAN COMMITTEE
ON ANTIMICROBIAL
SUSCEPTIBILITY TESTING

European Society of Clinical Microbiology and Infectious Diseases

- Tested concentrations ranged from 0.008 – 64 µg/mL
- WT-UL values for itraconazole, voriconazole, terbinafine and amorolfine

Antifungal susceptibility testing

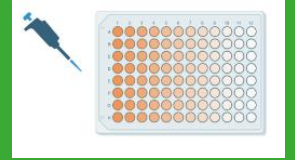


<i>T. indotineae</i> strains	TRB resistance	MIC (µg/mL)								
		FLC	ITC	VOR	KET	TRB	GRI	CPX	NAF	AMO
IHEM 28378	yes	16	0.032	0.25	0.25	16	8	1	32	0.25
IHEM 28379	yes	32	0.125	0.25	0.25	16	2	1	32	0.125
IHEM 28380	yes	64	0.25	0.25	0.5	16	4	1	32	0.125
IHEM 28384	yes	16	0,125	0,25	0,25	16	4	1	32	0,125
IHEM 28386	yes	32	0,125	0,25	0,25	16	4	0,5	32	0,125
IHEM 28387	yes	64	0,25	0,5	1	16	8	0,5	32	0,25
IHEM 28392	yes	64	0,25	0,25	0,25	16	4	0,5	32	0,125
IHEM 28394	yes	32	0,125	0,25	0,125	16	2	0,5	16	0,25
IHEM 28395	yes	32	0.125	0.25	0.25	16	4	0.5	32	0.125
IHEM 28381	no	64	0,25	0,25	0,5	0,016	2	1	0,25	0,032
IHEM 28382	no	32	0,125	0,125	0,5	0,064	2	1	0,125	0,064
IHEM 28383	no	64	0,25	1	1	0,064	4	1	0,25	0,064
IHEM 28385	no	32	0,5	0,5	0,5	0,032	4	0,5	0,125	0,064
IHEM 28388	no	8	0,125	0,125	0,125	0,064	2	1	0,064	0,064
IHEM 28389	no	16	0,125	0,125	0,125	0,125	2	1	0,125	0,125
IHEM 28390	no	64	0,25	0,5	0,5	0,125	4	1	0,25	0,125
IHEM 28391	no	64	0,25	0,5	0,5	0,032	4	0,5	0,25	0,064
IHEM 28393	no	16	0,064	0,125	0,125	0,064	8	0,5	0,5	0,25
IHEM 28396	no	32	0.25	0.25	0.5	0.032	2	0.5	0.25	0.25
IHEM 28397	no	16	0.064	0.125	0.125	0.032	2	0.5	0.064	0.125



Antifungal agent	GM MIC-values (µg/mL)		
	All (n = 20)	TRB susceptible (n = 11)	TRB resistant (n = 9)
Fluconazole	33.13	30.05	37.33
Itraconazole	0.16	0.17	0.15
Voriconazole	0.25	0.25	0.25
Ketoconazole	0.31	0.32	0.29
Terbinafine	0.67	0.05*	> 16.00*
Griseofulvin	3.48	2.92	4.32
Ciclopirox olamine	0.71	0.73	0.68
Naftifine	1.75	0.17*	29.63*
Amorolfine	0.12	0.09*	0.17*

Antifungal susceptibility testing



Antifungal agent	Geometric mean MIC-values (µg/mL)		
	All (n = 20)	TRB susceptible (n = 11)	TRB resistant (n = 9)
Fluconazole	33.13	30.05	37.33
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Ciclopirox olamine	0.71	0.73	0.68
Naftifine	1.75	0.17*	29.63*
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Nine out of twenty strains were determined **resistant** to TRB using WT-UL values by EUCAST.

All strains were considered **susceptible** for itraconazole, voriconazole and amorolfine.

Antifungal susceptibility testing



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Significant increase in the geometric mean of the MIC values between TRB susceptible and resistant strains for **naftifine** and **amorolfine**.

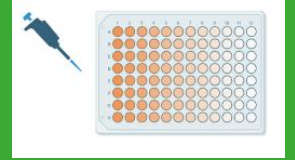
Naftifine and terbinafine are both **allylamines**, while amorolfine is a morpholine antifungal agent.

Potential of Ergosterol Synthesis Inhibitors To Cause Resistance or Cross-Resistance in *Trichophyton rubrum*

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Antifungal susceptibility testing



Antifungal agent	Geometric mean MIC-values (µg/mL)		
	All (n = 20)	TRB susceptible (n = 11)	TRB resistant (n = 9)
Fluconazole	33.13	30.05	37.33
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Ketoconazole	0.31	0.32	0.29
Terbinafine	0.67	0.05*	> 16.00*
Griseofulvin	3.48	2.92	4.32
Ciclopirox olamine	0.71	0.73	0.68
Naftifine	1.75	0.17*	29.63*
Amorolfine	0.12	0.09*	0.17*

Although MIC values are high for fluconazole, there are **no significantly elevated azole MIC** values between TRB susceptible and resistant strains for fluconazole or any of the other azoles.

DNA sequencing of *SQL*E



Terbinafine resistant strains

<i>T. indotineae</i> strains	TRB resistance	MIC (µg/mL)									<i>SQL</i> E missense mutations	amino acid substitution
		FLC	ITC	VOR	KET	TRB	GRI	CPX	NAF	AMO		
IHEM 28378	yes	16	0.032	0.25	0.25	16	8	1	32	0.25	T1189C	F397L
IHEM 28379	yes	32	0.125	0.25	0.25	16	2	1	32	0.125	T1189C	F397L
IHEM 28380	yes	64	0.25	0.25	0.5	16	4	1	32	0.125	T1189C	F397L
IHEM 28384	yes	16	0.125	0.25	0.25	16	4	1	32	0.125	T1189C	F397L
IHEM 28386	yes	32	0.125	0.25	0.25	16	4	0.5	32	0.125	C1191A	F397L
IHEM 28387	yes	64	0.25	0.5	1	16	8	0.5	32	0.25	T1189C	F397L
IHEM 28392	yes	64	0.25	0.25	0.25	16	4	0.5	32	0.125	C1191A	F397L
IHEM 28394	yes	32	0.125	0.25	0.125	16	2	0.5	16	0.25	A1179C	L393F
IHEM 28395	yes	32	0.125	0.25	0.25	16	4	0.5	32	0.125	T1189C & G1342A	F397L & A448T

Cross-resistance for naftifine?
Amorolfine?

TRB resistance

F397L
L393F
F397L & A448T

DNA sequencing of *SQLE*



Terbinafine susceptible strains

<i>T. indotineae</i> strains	TRB resistance	MIC (µg/mL)									<i>SQLE</i> missense mutations	amino acid substitution
		FLC	ITC	VOR	KET	TRB	GRI	CPX	NAF	AMO		
IHEM 28381	no	64	0.25	0,25	0.5	0.016	2	1	0.25	0.032	G1342A	A448T
IHEM 28382	no	32	0.125	0.125	0.5	0.064	2	1	0.125	0.064	none	none
IHEM 28383	no	64	0.25	1	1	0.064	4	1	0.25	0.064	G1342A	A448T
IHEM 28385	no	32	0.5	0.5	0.5	0.032	4	0.5	0.125	0.064	G1342A	A448T
IHEM 28388	no	8	0.125	0.125	0.125	0.064	2	1	0.064	0.064	none	none
IHEM 28389	no	16	0.125	0.125	0.125	0.125	2	1	0.125	0.125	G1342A	A448T
IHEM 28390	no	64	0.25	0.5	0.5	0.125	4	1	0.25	0.125	G1342A	A448T
IHEM 28391	no	64	0.25	0.5	0.5	0.032	4	0.5	0.25	0.064	G1342A	A448T
									0.5	0.25	G1342A	A448T
									0.25	0.25	G1342A	A448T
									0.064	0.125	G1342A & G1330A	A448T

Antifungal Susceptibility and Mutations in the Squalene Epoxidase Gene in Dermatophytes of the *Trichophyton mentagrophytes* Species Complex

Authors: [Xue Kong](#), [Chao Tang](#), [Ashutosh Singh](#), [Sarah A. Ahmed](#), [Abdullah M. S. Al-Hatmi](#), [Anuradha Chowdhary](#), [Pietro Nenoff](#)

No TRB resistance
No elevated azole MIC

A448T

Evaluation of MALDI-TOF MS as an identification tool



- Protein extraction based on protocol described by Cassagne *et al.* (2011)
- Spectra were compared to two databases:

①

**in-house database
BCCM/IHEM Sciensano**

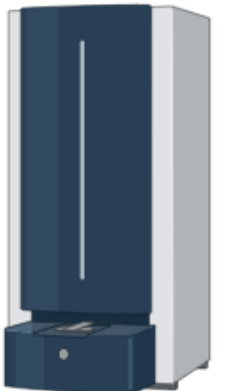


②

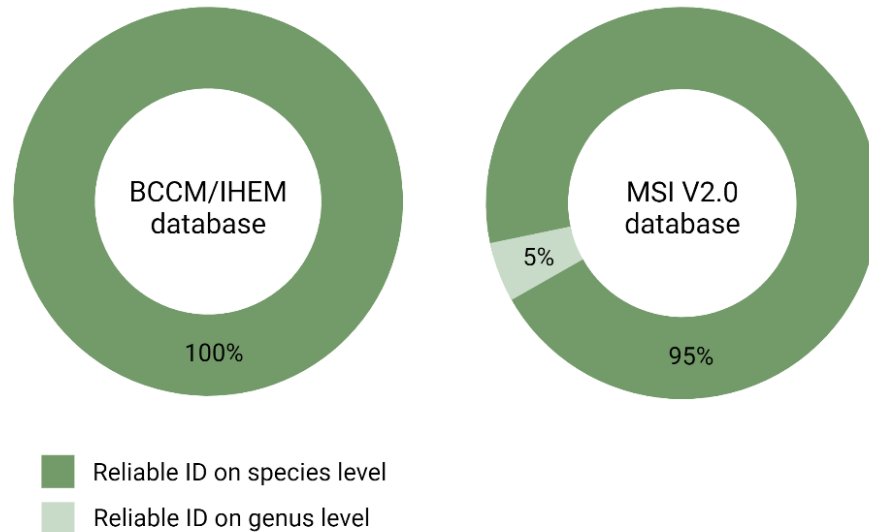
MSI V2.0 database
<https://msi.happy-dev.fr/>



- Can MALDI-TOF MS differentiate TRB resistant and susceptible strains?
Do susceptible strains have a susceptible reference strain as a best-match and vice versa?

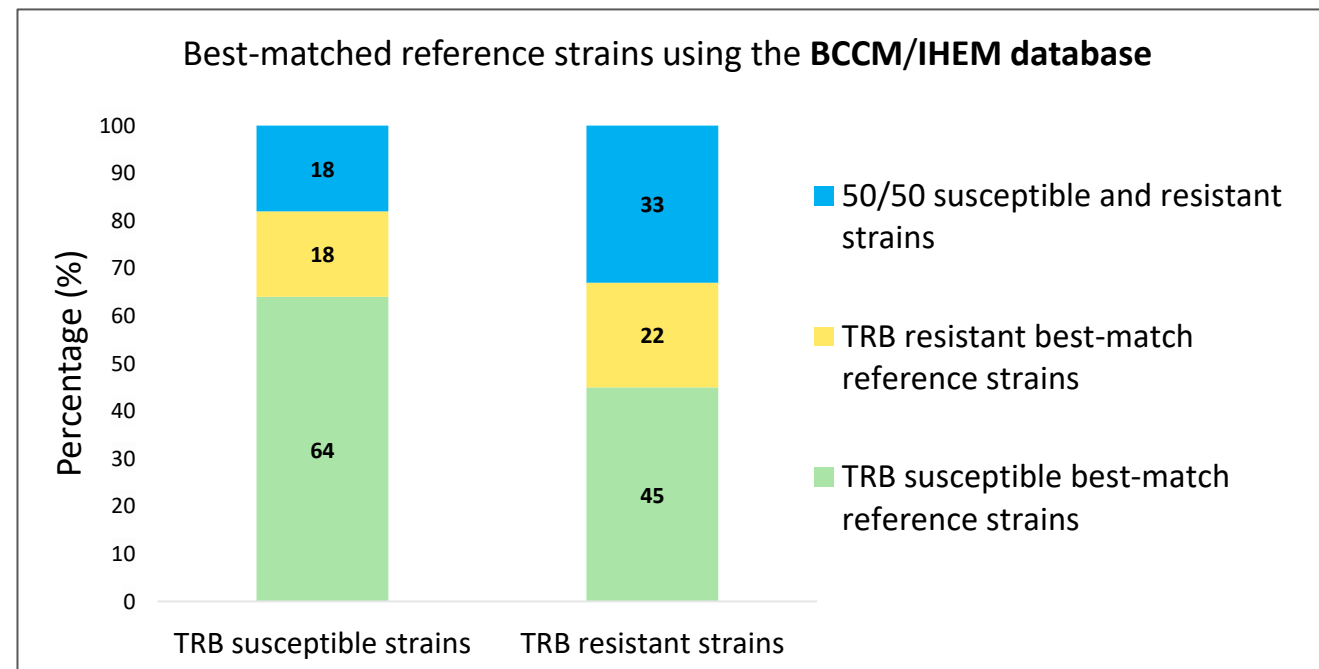


Evaluation of MALDI-TOF MS as an identification tool



It could **not differentiate** between TRB susceptible and resistant *T. indotineae* strains.

MALDI-TOF MS can **reliably distinguish** *T. indotineae* strains from closely related species, using both an in-house and public spectral database.



Conclusions

- TRB resistant *T. indotineae* strains showed significantly elevated MIC values for **naftifine** and **amorolfine**, raising suspicions of cross-resistance for naftifine.
- The observed TRB resistance was caused by point mutations in *SQLE*, resulting in amino acid substitutions **F397L** and **L393F**, while amino acid substitution **A448T** does not seem to cause **TRB** nor **azole resistance** when no other substitutions are present.
- MALDI-TOF MS is a **reliable tool for the identification** of *T. indotineae* strains, as it is able to distinguish them from closely related species *T. mentagrophytes* and *T. interdigitale*.
- MALDI-TOF MS could **not distinguish** TRB resistant *T. indotineae* strains from susceptible ones.

Acknowledgements & disclosures



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Dr Silke Urhlaß



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Dr Renaud Piarroux

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